CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SOARES DAIRY FARMS INC. – BADGER FLAT DAIRY

14515 S BADGER FLAT RD

LOS BANOS CA 93635

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Soares Dairy Farms, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
14515 S Badger Flat RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/01/1978

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X081-X080-X020-XXXX

B. OPERATORS

Soares, David Joseph

Operator name: Soares, David Joseph

Telephone no.: (209) 826-3414 Landline, (209) 752-9301 Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 S Badger Flat RD	Los Banos	CA	93635

Soares Dairy Farms Inc.

Operator name: Soares Dairy Farms Inc.

Telephone no.: (209) 826-3414 Landline, Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 S Badger Flat Rd	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

Soares, David Joseph

Legal owner name: Soares, David Joseph

Telephone no.: (209) 826-3414 Landline, (209) 752-9301 Cellular

Mailing Address Number and Street	City	State	Zip Code
14515 S Badger Flat RD	Los Banos	CA	93635

Soares Dairy Farms Inc.

Legal owner name: Soares Dairy Farms Inc.

Telephone no.: (209) 826-3414

Landline	Cellular

14515 S Badger Flat Rd	Los Banos	CA	93635
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Soares Living Trust

Legal owner name: Soares Living Trust

Telephone no.: (209) 826-3626

Landline	Cellular

14515 S Badger Flat RD	Los Banos	CA	93635
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	228	132	104	86
Number under roof	675	110	0	0	0	0
Maximum number	675	110	228	132	104	86
Average number	675	110	228	132	104	86
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,582.55 *tons per reporting period*

Total nitrogen from manure: 293,436.85 *lbs per reporting period*

After ammonia losses (30% loss applied): 205,405.80 *lbs per reporting period*

Total phosphorus from manure: 48,014.45 *lbs per reporting period*

Total potassium from manure: 130,599.15 *lbs per reporting period*

Total salt from manure: 343,118.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,907,000 *gallons*

Total nitrogen generated: 42,664.07 *lbs*

Total phosphorus generated: 6,527.65 *lbs*

Total potassium generated: 66,290.83 *lbs*

Total salt generated: 341,268.86 *lbs*

	11,907,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,907,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Drain Water	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
BF-1	11	11	2	process wastewater	X081-X080-X020-XXXX
BF-2	121	121	2	process wastewater	X081-X050-X006-XXXX X081-X080-X020-XXXX
BF-3	67	67	1	none	X081-X080-X022-XXXX X081-X080-X023-XXXX
BF-4	60	60	1	none	X081-X050-X008-XXXX
BF-5	16	16	1	manure	X081-X040-X006-XXXX
BF-6	15	15	0	none	X081-X040-X006-XXXX
BF-7	46	46	1	manure	X081-X040-X006-XXXX
BFH-1	36	36	1	manure	X063-X260-X012-XXXX
BFH-2	38	38	1	manure	X063-X260-X016-XXXX
BFH-3	117	117	1	manure	X063-X260-X015-XXXX X063-X260-X016-XXXX X063-X260-X022-XXXX X063-X260-X027-XXXX X063-X260-X028-XXXX
BFH-4A	51	51	2	manure	X063-X260-X028-XXXX
BFH-4B	51	51	2	manure	X063-X260-X028-XXXX
BFH-5	116	116	2	manure	X063-X260-X028-XXXX
BFHM-1	21	21	2	manure	X073-X160-X003-XXXX
BFHM-2	27	27	1	manure	X073-X160-X003-XXXX
BFHM-3	92	92	2	manure	X073-X160-X003-XXXX
Totals for areas that were used for application	743	743	20		
Totals for areas that were not used for application	142	142	2		
Land application area totals	885	885	22		

B. CROPS AND HARVESTS

BF-1

Field name: BF-1

BF-1

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	140.00 *ton*	Dry-weight		53.7	7,200.00	2,000.00	13,800.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.73	84.86	23.57	162.64	1,131.40

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	294.00 *ton*	Dry-weight		74.0	12,600.00	2,700.00	15,600.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.73	175.12	37.53	216.81	1,167.45

BF-2

Field name: BF-2

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 121 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	1,505.00 *ton*	Dry-weight		53.9	8,600.00	1,700.00	13,500.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.44	98.62	19.50	154.82	1,020.64

BF-2

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 121 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	3,180.00 *ton*	Dry-weight		68.6	17,300.00	2,300.00	22,300.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.28	285.53	37.96	368.05	1,336.86

BF-3

Field name: BF-3

10/29/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 67 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	492.00 *ton*	Dry-weight		58.9	32,000.00	2,900.00	33,800.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.34	193.16	17.50	204.02	676.05

BF-4

Field name: BF-4

BF-4

10/10/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 60 Plant date: 10/10/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	443.00 *ton*	Dry-weight		72.0	34,200.00	4,000.00	35,600.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.38	141.41	16.54	147.19	450.68

BF-5

Field name: BF-5

10/20/2017: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 16 Plant date: 10/20/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	119.00 *ton*	Dry-weight		64.9	23,600.00	4,200.00	22,400.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.44	123.22	21.93	116.95	522.11

BF-7

Field name: BF-7

BF-7

10/20/2017: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 46 Plant date: 10/20/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	344.00 *ton*	Dry-weight		71.9	31,400.00	4,600.00	32,000.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.48	131.97	19.33	134.49	458.10

BFH-1

Field name: BFH-1

11/18/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 36 Plant date: 11/18/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	260.00 *ton*	Dry-weight		21.0	34,700.00	3,350.00	19,850.00		8.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.22	395.97	38.23	226.51	952.83

BFH-2

Field name: BFH-2

BFH-2

11/18/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 38 Plant date: 11/18/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	277.00 *ton*	Dry-weight		23.3	33,450.00	3,000.00	16,400.00		7.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.29	374.04	33.55	183.39	821.88

BFH-3

Field name: BFH-3

10/15/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 117 Plant date: 10/15/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	848.00 *ton*	Dry-weight		21.5	33,200.00	2,850.00	17,600.00		7.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.25	377.79	32.43	200.27	881.88

BFH-4A

Field name: BFH-4A

BFH-4A

11/07/2022: Oats, hay

Crop: Oats, hay Acres planted: 51 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	204.00 *ton*	Dry-weight		7.8	9,100.00	2,200.00	13,800.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	67.12	16.23	101.79	516.32

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	1,335.00 *ton*	Dry-weight		67.5	9,300.00	2,800.00	10,800.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.18	158.24	47.64	183.76	884.76

BFH-4B

Field name: BFH-4B

11/22/2022: Oats, hay

Crop: Oats, hay Acres planted: 51 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	204.00 *ton*	Dry-weight		10.9	11,300.00	2,300.00	15,300.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	80.55	16.39	109.06	513.22

BFH-4B

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	1,335.00 *ton*	Dry-weight		67.5	9,300.00	2,800.00	10,800.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.18	158.24	47.64	183.76	884.76

BFH-5

Field name: BFH-5

10/28/2022: Oats, hay

Crop: Oats, hay Acres planted: 116 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	464.00 *ton*	Dry-weight		7.5	9,350.00	2,350.00	17,600.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	69.19	17.39	130.24	540.20

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 116 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	3,022.00 *ton*	Dry-weight		70.8	10,800.00	3,300.00	10,600.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.05	164.31	50.21	161.27	943.28

BFHM-1

Field name: BFHM-1

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	243.00 *ton*	Dry-weight		51.1	8,400.00	2,300.00	10,100.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.57	95.06	26.03	114.30	1,007.20

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	565.00 *ton*	Dry-weight		67.0	9,600.00	2,900.00	8,900.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.90	170.47	51.50	158.04	852.34

BFHM-2

Field name: BFHM-2

10/30/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 27 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	198.00 *ton*	Dry-weight		5.9	29,000.00	3,300.00	17,300.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.33	400.24	45.54	238.76	1,338.73

BFHM-3

Field name: BFHM-3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 92 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	1,027.00 *ton*	Dry-weight		52.5	10,400.00	2,600.00	16,500.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.16	110.29	27.57	174.98	1,293.80

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/07/2023	2,399.00 *ton*	Dry-weight		68.4	10,100.00	2,800.00	8,900.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.08	166.45	46.14	146.67	906.40

NUTRIENT BUDGET

A. LAND APPLICATIONS

BF-1 - 10/29/2022: Oats, silage-soft dough

Field name: BF-1

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.99	9.59	197.62	958.61	648,000.00 *gal*
Canal	Surface water	1.00	0.00	0.00	286.80	1,890,220.00 *gal*
Application event totals		65.99	9.59	197.62	1,245.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	38.19	0.00	0.00	0.00	1,530,000.00 *gal*
Application event totals		38.19	0.00	0.00	0.00	

BF-1 - 06/03/2023: Corn, silage

Field name: BF-1

Crop: Corn, silage — Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	33.69	0.00	0.00	0.00	1,350,000.00 *gal*
Application event totals		33.69	0.00	0.00	0.00	

BF-1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.01	10.72	85.29	470.70	162,000.00 *gal*
Canal	Surface water	1.63	0.00	0.00	464.81	3,063,460.00 *gal*
Application event totals		66.64	10.72	85.29	935.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.84	9.67	103.11	510.03	162,000.00 *gal*
Irrigation Well	Ground water	37.44	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		105.28	9.67	103.11	510.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	370.86	2,444,250.00 *gal*
Application event totals		1.30	0.00	0.00	370.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	26.96	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		26.96	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	405.47	2,672,380.00 *gal*
Application event totals		1.42	0.00	0.00	405.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	405.47	2,672,380.00 *gal*
Application event totals		1.42	0.00	0.00	405.47	

BF-2 - 10/30/2022: Oats, silage-soft dough

Field name: BF-2

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.63	3.49	71.86	348.59	2,592,000.00 *gal*
Canal	Surface water	0.57	0.00	0.00	161.83	11,732,400.00 *gal*
Application event totals		24.20	3.49	71.86	510.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.80	12.99	103.38	570.55	2,160,000.00 *gal*
Irrigation Well	Ground water	27.23	0.00	0.00	0.00	12,000,000.00 *gal*
Application event totals		106.03	12.99	103.38	570.55	

BF-2 - 06/04/2023: Corn, silage

Field name: BF-2

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	33.69	0.00	0.00	0.00	14,850,000.00 *gal*
Application event totals		33.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.88	12.18	96.92	534.89	2,025,000.00 *gal*
Canal	Surface water	1.63	0.00	0.00	464.81	33,698,060.00 *gal*
Application event totals		75.51	12.18	96.92	999.70	

BF-2 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	37.44	0.00	0.00	0.00	16,500,000.00 *gal*
Application event totals		37.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.17	11.43	121.86	602.76	2,106,000.00 *gal*
Canal	Surface water	1.30	0.00	0.00	370.86	26,886,750.00 *gal*
Application event totals		81.47	11.43	121.86	973.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.12	11.14	118.74	587.31	2,052,000.00 *gal*
Irrigation Well	Ground water	26.96	0.00	0.00	0.00	11,880,000.00 *gal*
Application event totals		105.07	11.14	118.74	587.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	405.47	29,396,180.00 *gal*
Application event totals		1.42	0.00	0.00	405.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	405.47	29,396,180.00 *gal*
Application event totals		1.42	0.00	0.00	405.47	

BF-3 - 10/29/2022: Alfalfa, hay

Field name: BF-3

Crop: Alfalfa, hay

Plant date: 10/29/2022

BF-3 - 10/29/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	371.82	14,926,220.00 *gal*
Application event totals		1.30	0.00	0.00	371.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	313.37	12,579,740.00 *gal*
Application event totals		1.10	0.00	0.00	313.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.79	0.00	0.00	511.45	20,531,700.00 *gal*
Application event totals		1.79	0.00	0.00	511.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	362.89	14,567,730.00 *gal*
Application event totals		1.27	0.00	0.00	362.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.33	0.00	0.00	380.75	15,284,710.00 *gal*
Application event totals		1.33	0.00	0.00	380.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	389.68	15,643,200.00 *gal*
Application event totals		1.36	0.00	0.00	389.68	

BF-4 - 10/10/2021: Alfalfa, hay

BF-4 - 10/10/2021: Alfalfa, hay

Field name: BF-4

Crop: Alfalfa, hay

Plant date: 10/10/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	487.72	17,533,420.00 *gal*
Application event totals		1.71	0.00	0.00	487.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.43	0.00	0.00	409.76	14,730,680.00 *gal*
Application event totals		1.43	0.00	0.00	409.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	335.42	12,058,300.00 *gal*
Application event totals		1.17	0.00	0.00	335.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.65	0.00	0.00	472.31	16,979,390.00 *gal*
Application event totals		1.65	0.00	0.00	472.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.74	0.00	0.00	495.88	17,826,730.00 *gal*
Application event totals		1.74	0.00	0.00	495.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.89	0.00	0.00	253.83	9,125,200.00 *gal*
Application event totals		0.89	0.00	0.00	253.83	

BF-5 - 10/20/2017: Alfalfa, hay

Field name: BF-5

Crop: Alfalfa, hay

Plant date: 10/20/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.51	0.00	0.00	431.74	4,138,930.00 *gal*
Application event totals		1.51	0.00	0.00	431.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	363.75	3,487,130.00 *gal*
Application event totals		1.27	0.00	0.00	363.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	299.16	2,867,920.00 *gal*
Application event totals		1.05	0.00	0.00	299.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.42	0.00	0.00	404.55	3,878,210.00 *gal*
Application event totals		1.42	0.00	0.00	404.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.57	0.00	0.00	448.74	4,301,880.00 *gal*
Application event totals		1.57	0.00	0.00	448.74	

BF-5 - 10/20/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.80	0.00	0.00	227.77	2,183,530.00 *gal*
Application event totals		0.80	0.00	0.00	227.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	148.62	24.26	114.24	6,177.21	120.00 *ton*
Application event totals		148.62	24.26	114.24	6,177.21	

BF-7 - 10/20/2017: Alfalfa, hay

Field name: BF-7

Crop: Alfalfa, hay

Plant date: 10/20/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	422.13	11,634,630.00 *gal*
Application event totals		1.48	0.00	0.00	422.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	381.93	10,526,570.00 *gal*
Application event totals		1.34	0.00	0.00	381.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.54	0.00	0.00	439.87	12,123,480.00 *gal*
Application event totals		1.54	0.00	0.00	439.87	

BF-7 - 10/20/2017: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	300.34	8,277,860.00 *gal*
Application event totals		1.05	0.00	0.00	300.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.12	0.00	0.00	320.44	8,831,890.00 *gal*
Application event totals		1.12	0.00	0.00	320.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	374.84	10,331,030.00 *gal*
Application event totals		1.31	0.00	0.00	374.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	161.54	26.37	124.18	6,714.36	375.00 *ton*
Application event totals		161.54	26.37	124.18	6,714.36	

BFH-1 - 11/18/2021: Alfalfa, hay

Field name: BFH-1

Crop: Alfalfa, hay

Plant date: 11/18/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	262.90	5,670,660.00 *gal*
Application event totals		0.92	0.00	0.00	262.90	

BFH-1 - 11/18/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.61	0.00	0.00	459.32	9,907,360.00 *gal*
Application event totals		1.61	0.00	0.00	459.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.63	0.00	0.00	465.36	10,037,720.00 *gal*
Application event totals		1.63	0.00	0.00	465.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.73	0.00	0.00	494.07	10,656,930.00 *gal*
Application event totals		1.73	0.00	0.00	494.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.61	0.00	0.00	459.32	9,907,360.00 *gal*
Application event totals		1.61	0.00	0.00	459.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.86	0.00	0.00	244.77	5,279,580.00 *gal*
Application event totals		0.86	0.00	0.00	244.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	257.60	42.06	198.02	10,707.16	468.00 *ton*
Application event totals		257.60	42.06	198.02	10,707.16	

BFH-2 - 11/18/2021: Alfalfa, hay

BFH-2 - 11/18/2021: Alfalfa, hay

Field name: BFH-2

Crop: Alfalfa, hay

Plant date: 11/18/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.16	0.00	0.00	332.08	7,560,880.00 *gal*
Application event totals		1.16	0.00	0.00	332.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.64	0.00	0.00	0.00	7,020,000.00 *gal*
Application event totals		12.64	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	309.18	7,039,440.00 *gal*
Application event totals		1.08	0.00	0.00	309.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.16	0.00	0.00	0.00	6,750,000.00 *gal*
Application event totals		12.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	310.61	7,072,030.00 *gal*
Application event totals		1.09	0.00	0.00	310.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.48	0.00	0.00	0.00	6,930,000.00 *gal*
Application event totals		12.48	0.00	0.00	0.00	

BFH-2 - 11/18/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	286.28	6,518,000.00 *gal*
Application event totals		1.00	0.00	0.00	286.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	309.18	7,039,440.00 *gal*
Application event totals		1.08	0.00	0.00	309.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	257.60	42.06	198.02	10,707.16	494.00 *ton*
Application event totals		257.60	42.06	198.02	10,707.16	

BFH-3 - 10/15/2019: Alfalfa, hay

Field name: BFH-3

Crop: Alfalfa, hay

Plant date: 10/15/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.15	0.00	0.00	327.29	22,943,360.00 *gal*
Application event totals		1.15	0.00	0.00	327.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.90	0.00	0.00	0.00	13,500,000.00 *gal*
Application event totals		7.90	0.00	0.00	0.00	

BFH-3 - 10/15/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	312.41	21,900,480.00 *gal*
Application event totals		1.09	0.00	0.00	312.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	374.47	26,251,245.00 *gal*
Application event totals		1.31	0.00	0.00	374.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.18	0.00	0.00	337.51	23,660,340.00 *gal*
Application event totals		1.18	0.00	0.00	337.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.37	0.00	0.00	0.00	12,600,000.00 *gal*
Application event totals		7.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.09	0.00	0.00	311.94	21,867,890.00 *gal*
Application event totals		1.09	0.00	0.00	311.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.59	0.00	0.00	167.83	11,764,990.00 *gal*
Application event totals		0.59	0.00	0.00	167.83	

BFH-3 - 10/15/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.63	0.00	0.00	0.00	13,050,000.00 *gal*
Application event totals		7.63	0.00	0.00	0.00	

10/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	257.60	42.06	198.02	10,707.16	1,521.00 *ton*
Application event totals		257.60	42.06	198.02	10,707.16	

BFH-4A - 11/07/2022: Oats, hay

Field name: BFH-4A

Crop: Oats, hay

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	65.70	10.86	22.81	0.00	230.00 *ton*
Application event totals		65.70	10.86	22.81	0.00	

11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.89	0.00	0.00	0.00	7,369,266.00 *gal*
Application event totals		9.89	0.00	0.00	0.00	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.53	0.00	0.00	0.00	8,590,459.00 *gal*
Application event totals		11.53	0.00	0.00	0.00	

BFH-4A - 06/04/2023: Corn, silage

Field name: BFH-4A

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	159.30	26.01	122.46	6,621.32	410.00 *ton*
Application event totals		159.30	26.01	122.46	6,621.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.44	0.00	0.00	0.00	8,527,294.00 *gal*
Application event totals		11.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	265.23	8,104,595.00 *gal*
Application event totals		0.93	0.00	0.00	265.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.54	0.00	0.00	0.00	7,853,532.00 *gal*
Application event totals		10.54	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.72	0.00	0.00	206.09	6,297,644.00 *gal*
Application event totals		0.72	0.00	0.00	206.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.66	0.00	0.00	188.63	5,763,945.00 *gal*
Application event totals		0.66	0.00	0.00	188.63	

BFH-4A - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.45	0.00	0.00	0.00	7,790,367.00 *gal*
Application event totals		10.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.62	0.00	0.00	175.90	5,375,108.00 *gal*
Application event totals		0.62	0.00	0.00	175.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.14	0.00	0.00	0.00	7,558,761.00 *gal*
Application event totals		10.14	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.45	0.00	0.00	0.00	7,790,367.00 *gal*
Application event totals		10.45	0.00	0.00	0.00	

BFH-4B - 11/22/2022: Oats, hay

Field name: BFH-4B

Crop: Oats, hay

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	65.70	10.86	22.81	0.00	230.00 *ton*
Application event totals		65.70	10.86	22.81	0.00	

BFH-4B - 11/22/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.89	0.00	0.00	0.00	7,369,266.00 *gal*
Application event totals		9.89	0.00	0.00	0.00	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.53	0.00	0.00	0.00	8,590,459.00 *gal*
Application event totals		11.53	0.00	0.00	0.00	

BFH-4B - 06/04/2023: Corn, silage

Field name: BFH-4B

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	159.30	26.01	122.46	6,621.32	410.00 *ton*
Application event totals		159.30	26.01	122.46	6,621.32	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.44	0.00	0.00	0.00	8,527,294.00 *gal*
Application event totals		11.44	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	265.23	8,104,595.00 *gal*
Application event totals		0.93	0.00	0.00	265.23	

BFH-4B - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.90	0.00	0.00	257.01	7,853,532.00 *gal*
Application event totals		0.90	0.00	0.00	257.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.45	0.00	0.00	0.00	6,297,644.00 *gal*
Application event totals		8.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.66	0.00	0.00	188.63	5,763,945.00 *gal*
Application event totals		0.66	0.00	0.00	188.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.45	0.00	0.00	0.00	7,790,367.00 *gal*
Application event totals		10.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.72	0.00	0.00	205.22	6,270,959.00 *gal*
Application event totals		0.72	0.00	0.00	205.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.14	0.00	0.00	0.00	7,558,761.00 *gal*
Application event totals		10.14	0.00	0.00	0.00	

BFH-4B - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.45	0.00	0.00	0.00	7,790,367.00 *gal*
Application event totals		10.45	0.00	0.00	0.00	

BFH-5 - 10/28/2022: Oats, hay

Field name: BFH-5

Crop: Oats, hay

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	68.45	11.31	23.76	0.00	545.00 *ton*
Application event totals		68.45	11.31	23.76	0.00	

11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.89	0.00	0.00	0.00	16,761,468.00 *gal*
Application event totals		9.89	0.00	0.00	0.00	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.53	0.00	0.00	0.00	19,539,083.00 *gal*
Application event totals		11.53	0.00	0.00	0.00	

BFH-5 - 06/04/2023: Corn, silage

Field name: BFH-5

Crop: Corn, silage

Plant date: 06/04/2023

BFH-5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	163.99	26.77	126.06	6,816.23	960.00 *ton*
Application event totals		163.99	26.77	126.06	6,816.23	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.44	0.00	0.00	0.00	19,395,413.00 *gal*
Application event totals		11.44	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.93	0.00	0.00	265.23	18,433,980.00 *gal*
Application event totals		0.93	0.00	0.00	265.23	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	294.42	20,462,932.00 *gal*
Application event totals		1.03	0.00	0.00	294.42	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.74	0.00	0.00	0.00	18,198,165.00 *gal*
Application event totals		10.74	0.00	0.00	0.00	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.75	0.00	0.00	215.57	14,983,028.00 *gal*
Application event totals		0.75	0.00	0.00	215.57	

BFH-5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.83	0.00	0.00	0.00	23,442,110.00 *gal*
Application event totals		13.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.21	0.00	0.00	58.63	4,075,245.00 *gal*
Application event totals		0.21	0.00	0.00	58.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.17	0.00	0.00	0.00	17,240,367.00 *gal*
Application event totals		10.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.71	0.00	0.00	0.00	18,150,275.00 *gal*
Application event totals		10.71	0.00	0.00	0.00	

BFHM-1 - 10/29/2022: Oats, silage-soft dough

Field name: BFHM-1

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	114.47	18.92	39.73	0.00	165.00 *ton*
Application event totals		114.47	18.92	39.73	0.00	

BFHM-1 - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.49	0.00	0.00	364.94	3,222,324.00 *gal*
Application event totals		4.49	0.00	0.00	364.94	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.20	0.00	0.00	340.93	3,010,329.00 *gal*
Application event totals		4.20	0.00	0.00	340.93	

BFHM-1 - 06/03/2023: Corn, silage

Field name: BFHM-1

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	188.72	30.81	145.07	7,844.08	200.00 *ton*
Application event totals		188.72	30.81	145.07	7,844.08	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	9.63	0.00	0.00	782.01	6,904,980.00 *gal*
Application event totals		9.63	0.00	0.00	782.01	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.15	0.00	0.00	418.45	3,694,770.00 *gal*
Application event totals		5.15	0.00	0.00	418.45	

BFHM-1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.16	0.00	0.00	419.13	3,700,827.00 *gal*
Application event totals		5.16	0.00	0.00	419.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.93	0.00	0.00	399.92	3,531,231.00 *gal*
Application event totals		4.93	0.00	0.00	399.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.29	0.00	0.00	348.48	3,076,956.00 *gal*
Application event totals		4.29	0.00	0.00	348.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.98	0.00	0.00	404.73	3,573,630.00 *gal*
Application event totals		4.98	0.00	0.00	404.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.25	0.00	0.00	425.99	3,761,397.00 *gal*
Application event totals		5.25	0.00	0.00	425.99	

BFHM-2 - 10/30/2022: Alfalfa, hay

Field name: BFHM-2

Crop: Alfalfa, hay

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

BFHM-2 - 10/30/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.20	0.00	0.00	340.93	3,870,423.00 *gal*
Application event totals		4.20	0.00	0.00	340.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.82	0.00	0.00	391.01	4,438,916.00 *gal*
Application event totals		4.82	0.00	0.00	391.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.15	0.00	0.00	418.45	4,750,419.00 *gal*
Application event totals		5.15	0.00	0.00	418.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.16	0.00	0.00	419.13	4,758,206.00 *gal*
Application event totals		5.16	0.00	0.00	419.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.93	0.00	0.00	399.92	4,540,154.00 *gal*
Application event totals		4.93	0.00	0.00	399.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.29	0.00	0.00	348.48	3,956,086.00 *gal*
Application event totals		4.29	0.00	0.00	348.48	

BFHM-2 - 10/30/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.98	0.00	0.00	404.73	4,594,667.00 *gal*
Application event totals		4.98	0.00	0.00	404.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.25	0.00	0.00	425.99	4,836,082.00 *gal*
Application event totals		5.25	0.00	0.00	425.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	257.60	42.06	198.02	10,707.16	351.00 *ton*
Application event totals		257.60	42.06	198.02	10,707.16	

BFHM-3 - 11/01/2022: Oats, silage-soft dough

Field name: BFHM-3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	137.77	22.77	47.82	0.00	870.00 *ton*
Application event totals		137.77	22.77	47.82	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.49	0.00	0.00	364.94	14,116,848.00 *gal*
Application event totals		4.49	0.00	0.00	364.94	

BFHM-3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.20	0.00	0.00	340.93	13,188,108.00 *gal*
Application event totals		4.20	0.00	0.00	340.93	

BFHM-3 - 06/04/2023: Corn, silage

Field name: BFHM-3

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	187.39	30.59	144.05	7,788.66	870.00 *ton*
Application event totals		187.39	30.59	144.05	7,788.66	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	9.63	0.00	0.00	782.01	30,250,389.00 *gal*
Application event totals		9.63	0.00	0.00	782.01	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.15	0.00	0.00	418.45	16,186,611.00 *gal*
Application event totals		5.15	0.00	0.00	418.45	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.16	0.00	0.00	419.13	16,213,147.00 *gal*
Application event totals		5.16	0.00	0.00	419.13	

BFHM-3 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.93	0.00	0.00	399.92	15,470,155.00 *gal*
Application event totals		4.93	0.00	0.00	399.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.29	0.00	0.00	348.48	13,479,998.00 *gal*
Application event totals		4.29	0.00	0.00	348.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	4.98	0.00	0.00	404.73	15,655,903.00 *gal*
Application event totals		4.98	0.00	0.00	404.73	

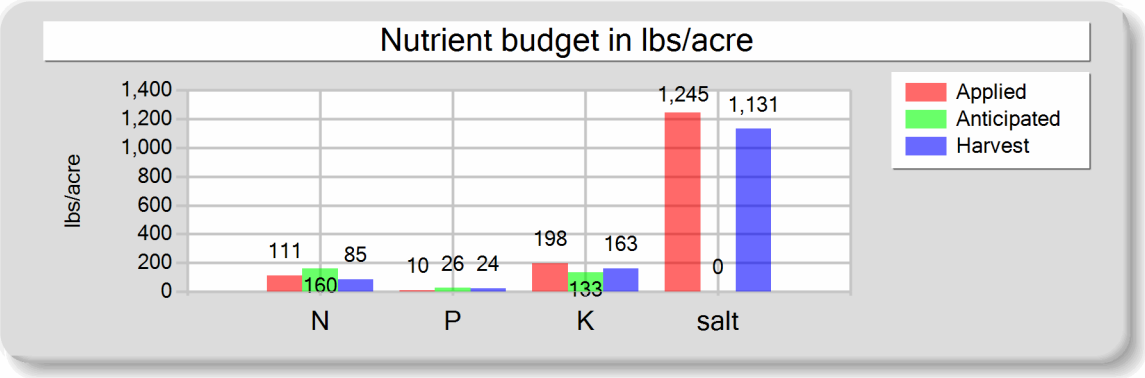
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Drain Water	Surface water	5.25	0.00	0.00	425.99	16,478,501.00 *gal*
Application event totals		5.25	0.00	0.00	425.99	

B. NUTRIENT BUDGET

BF-1 - 10/29/2022: Oats, silage-soft dough

Field name: BF-1 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	64.99	9.59	197.62	958.61
Fresh water	39.19	0.00	0.00	286.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.18	9.59	197.62	1,245.41
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	84.86	23.57	162.64	1,131.40
Nutrient balance	26.33	-13.98	34.98	114.01
Applied to removed ratio	1.31	0.41	1.22	1.10

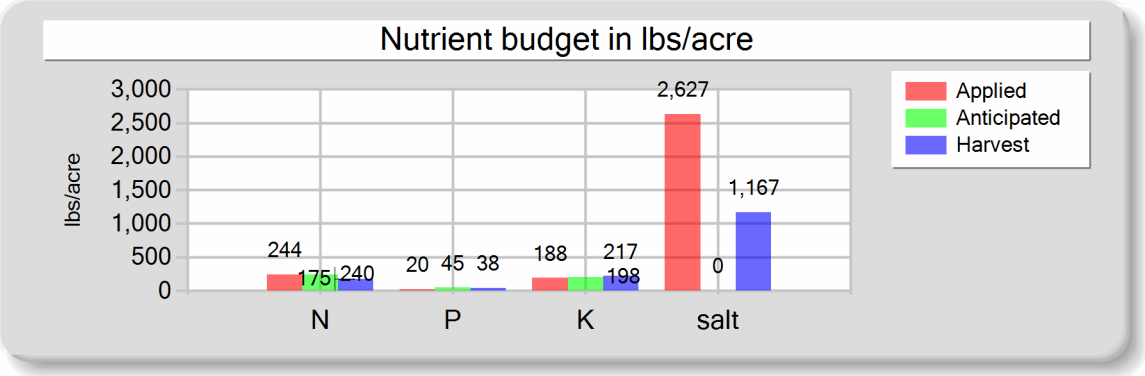
Fresh water applied	
3,420,220.00	*gallons*
125.96	*acre-inches*
11.45	*inches/acre*

Process wastewater applied	
648,000.00	*gallons*
23.86	*acre-inches*
2.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-1 - 06/03/2023: Corn, silage

Field name: BF-1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	132.85	20.39	188.40	980.73
Fresh water	103.85	0.00	0.00	1,646.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.71	20.39	188.40	2,627.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	175.12	37.53	216.81	1,167.45
Nutrient balance	68.59	-17.14	-28.41	1,459.90
Applied to removed ratio	1.39	0.54	0.87	2.25

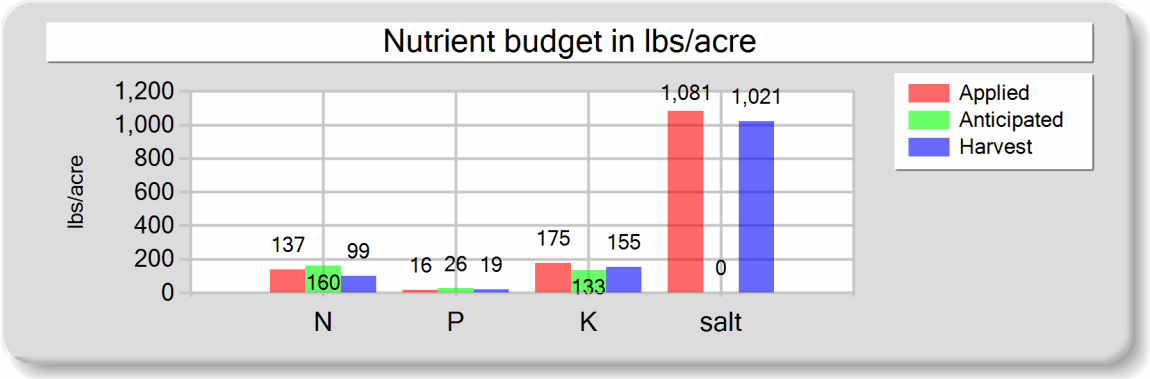
Fresh water applied	
14,782,470.00	*gallons*
544.39	*acre-inches*
49.49	*inches/acre*

Process wastewater applied	
324,000.00	*gallons*
11.93	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-2 - 10/30/2022: Oats, silage-soft dough

Field name: BF-2 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	102.44	16.48	175.25	919.14
Fresh water	27.79	0.00	0.00	161.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.23	16.48	175.25	1,080.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	98.62	19.50	154.82	1,020.64
Nutrient balance	38.61	-3.02	20.43	60.33
Applied to removed ratio	1.39	0.85	1.13	1.06

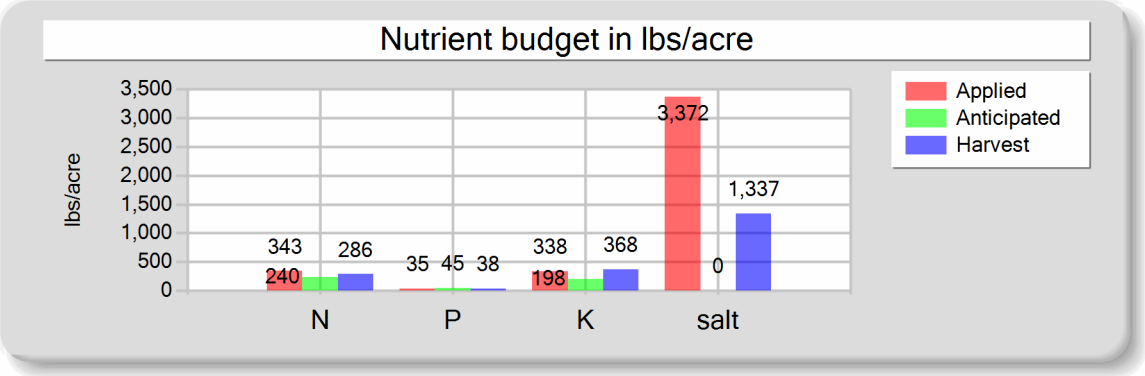
Fresh water applied	
23,732,400.00	*gallons*
873.98	*acre-inches*
7.22	*inches/acre*

Process wastewater applied	
4,752,000.00	*gallons*
175.00	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-2 - 06/04/2023: Corn, silage

Field name: BF-2 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.17	34.75	337.52	1,724.96
Fresh water	103.85	0.00	0.00	1,646.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.03	34.75	337.52	3,371.58
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	285.53	37.96	368.05	1,336.86
Nutrient balance	57.50	-3.21	-30.53	2,034.72
Applied to removed ratio	1.20	0.92	0.92	2.52

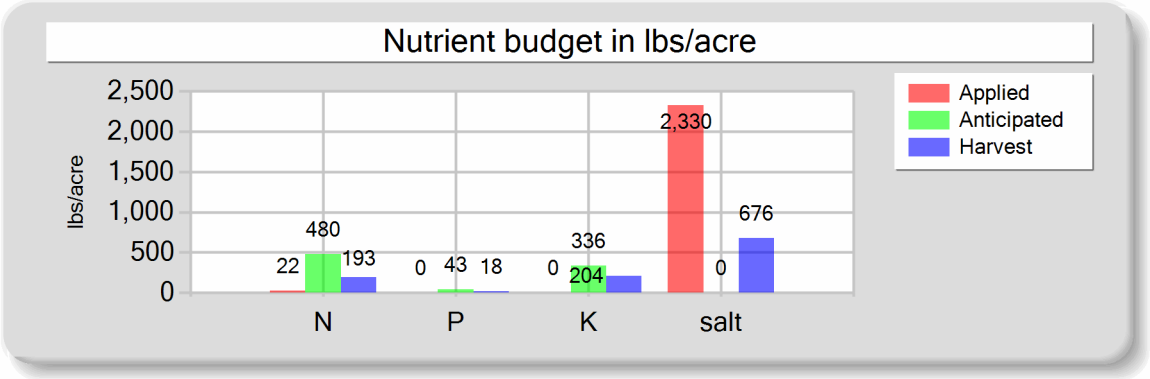
Fresh water applied	
162,607,170.00	*gallons*
5,988.27	*acre-inches*
49.49	*inches/acre*

Process wastewater applied	
6,183,000.00	*gallons*
227.70	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-3 - 10/29/2022: Alfalfa, hay

Field name: BF-3 Crop: Alfalfa, hay Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.15	0.00	0.00	2,329.96
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	22.15	0.00	0.00	2,329.96
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	193.16	17.50	204.02	676.05
Nutrient balance	-171.00	-17.50	-204.02	1,653.90
Applied to removed ratio	0.11	0.00	0.00	3.45

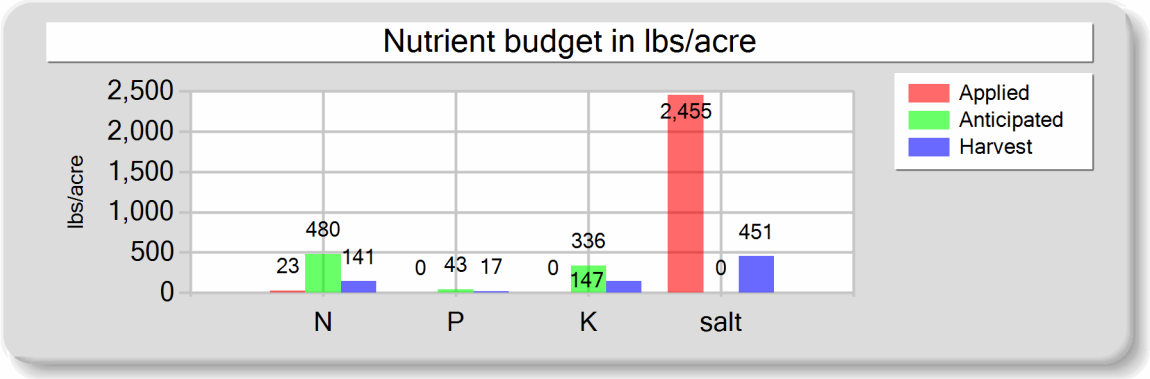
Fresh water applied	
93,533,300.00	*gallons*
3,444.51	*acre-inches*
51.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-4 - 10/10/2021: Alfalfa, hay

Field name: BF-4 Crop: Alfalfa, hay Plant date: 10/10/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.59	0.00	0.00	2,454.92
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	22.59	0.00	0.00	2,454.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	141.41	16.54	147.19	450.68
Nutrient balance	-118.81	-16.54	-147.19	2,004.25
Applied to removed ratio	0.16	0.00	0.00	5.45

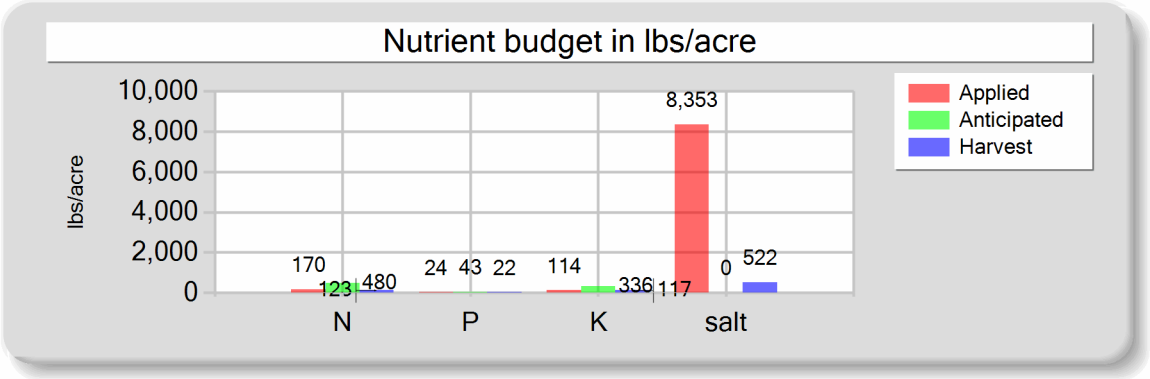
Fresh water applied	
88,253,720.00	*gallons*
3,250.08	*acre-inches*
54.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-5 - 10/20/2017: Alfalfa, hay

Field name: BF-5 Crop: Alfalfa, hay Plant date: 10/20/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	148.62	24.26	114.24	6,177.21
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.61	0.00	0.00	2,175.71
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	170.23	24.26	114.24	8,352.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	123.22	21.93	116.95	522.11
Nutrient balance	47.01	2.34	-2.71	7,830.81
Applied to removed ratio	1.38	1.11	0.98	16.00

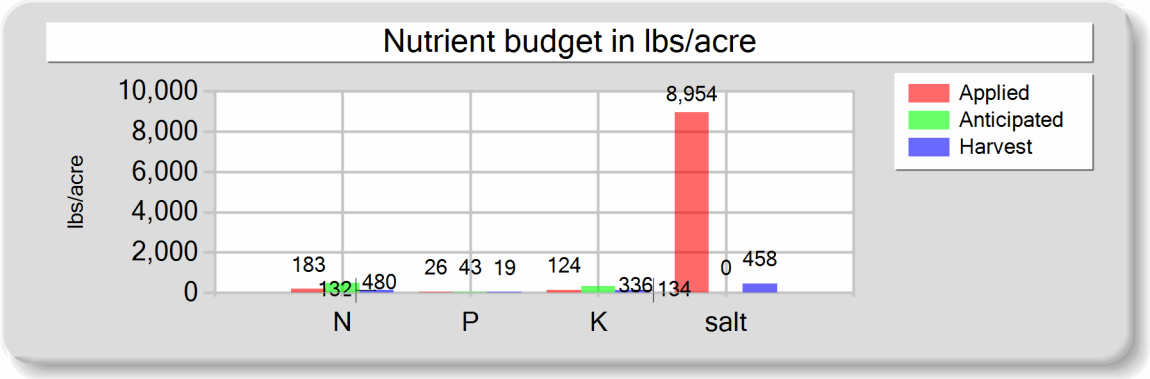
Fresh water applied	
20,857,600.00	*gallons*
768.11	*acre-inches*
48.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BF-7 - 10/20/2017: Alfalfa, hay

Field name: BF-7 Crop: Alfalfa, hay Plant date: 10/20/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	161.54	26.37	124.18	6,714.36
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.84	0.00	0.00	2,239.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	183.38	26.37	124.18	8,953.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	131.97	19.33	134.49	458.10
Nutrient balance	51.41	7.04	-10.31	8,495.82
Applied to removed ratio	1.39	1.36	0.92	19.55

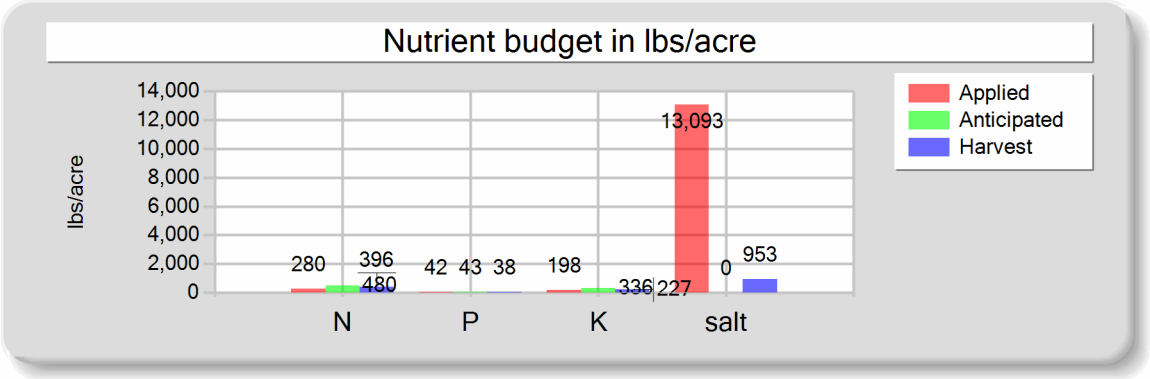
Fresh water applied	
61,725,460.00	*gallons*
2,273.14	*acre-inches*
49.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-1 - 11/18/2021: Alfalfa, hay

Field name: BFH-1 Crop: Alfalfa, hay Plant date: 11/18/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	257.60	42.06	198.02	10,707.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.35	0.00	0.00	2,385.72
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	279.95	42.06	198.02	13,092.89
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	395.97	38.23	226.51	952.83
Nutrient balance	-116.01	3.83	-28.49	12,140.06
Applied to removed ratio	0.71	1.10	0.87	13.74

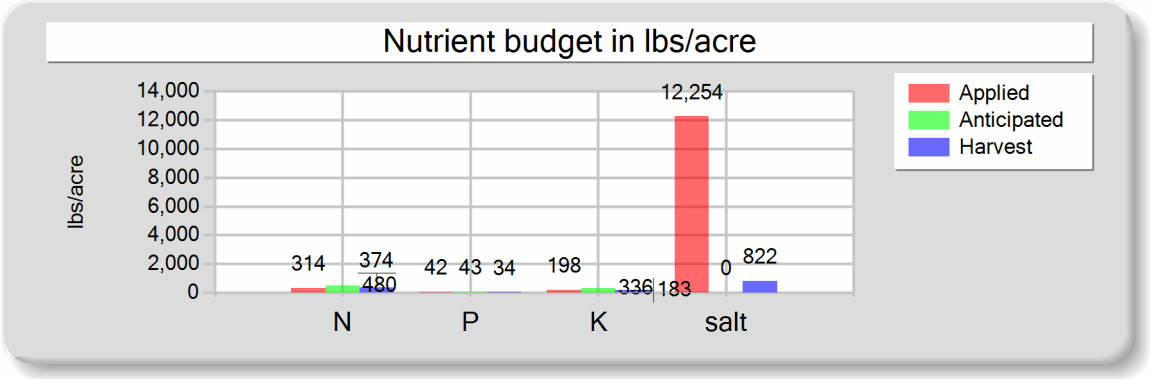
Fresh water applied	
51,459,610.00	*gallons*
1,895.08	*acre-inches*
52.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-2 - 11/18/2021: Alfalfa, hay

Field name: BFH-2 Crop: Alfalfa, hay Plant date: 11/18/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	257.60	42.06	198.02	10,707.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	42.69	0.00	0.00	1,547.33
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.29	42.06	198.02	12,254.49
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	374.04	33.55	183.39	821.88
Nutrient balance	-59.75	8.51	14.64	11,432.61
Applied to removed ratio	0.84	1.25	1.08	14.91

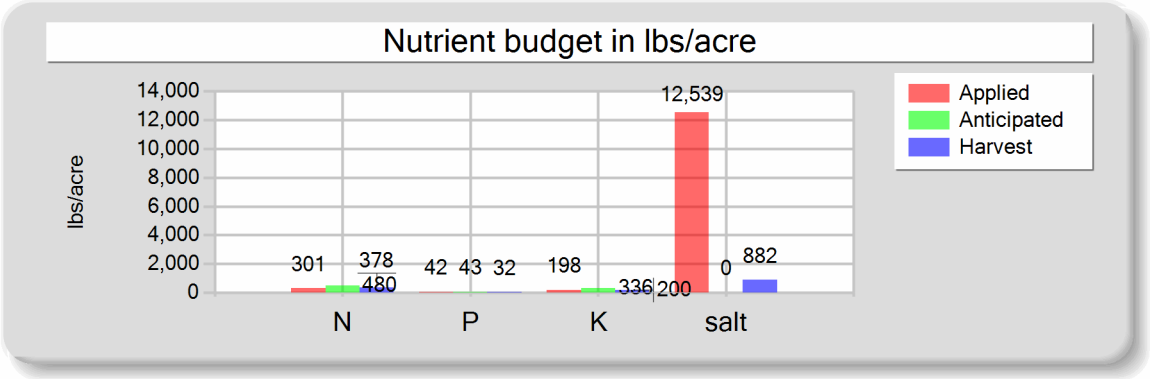
Fresh water applied	
55,929,790.00	*gallons*
2,059.70	*acre-inches*
54.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-3 - 10/15/2019: Alfalfa, hay

Field name: BFH-3 Crop: Alfalfa, hay Plant date: 10/15/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	257.60	42.06	198.02	10,707.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	29.31	0.00	0.00	1,831.45
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	300.91	42.06	198.02	12,538.62
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	377.79	32.43	200.27	881.88
Nutrient balance	-76.88	9.63	-2.25	11,656.73
Applied to removed ratio	0.80	1.30	0.99	14.22

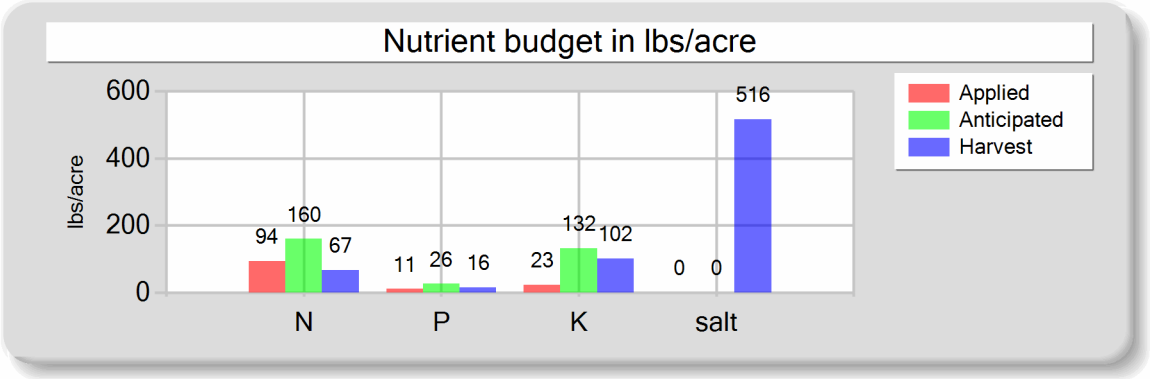
Fresh water applied	
167,538,305.00	*gallons*
6,169.87	*acre-inches*
52.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-4A - 11/07/2022: Oats, hay

Field name: BFH-4A Crop: Oats, hay Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	65.70	10.86	22.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	94.11	10.86	22.81	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	67.12	16.23	101.79	516.32
Nutrient balance	26.99	-5.37	-78.98	-516.32
Applied to removed ratio	1.40	0.67	0.22	0.00

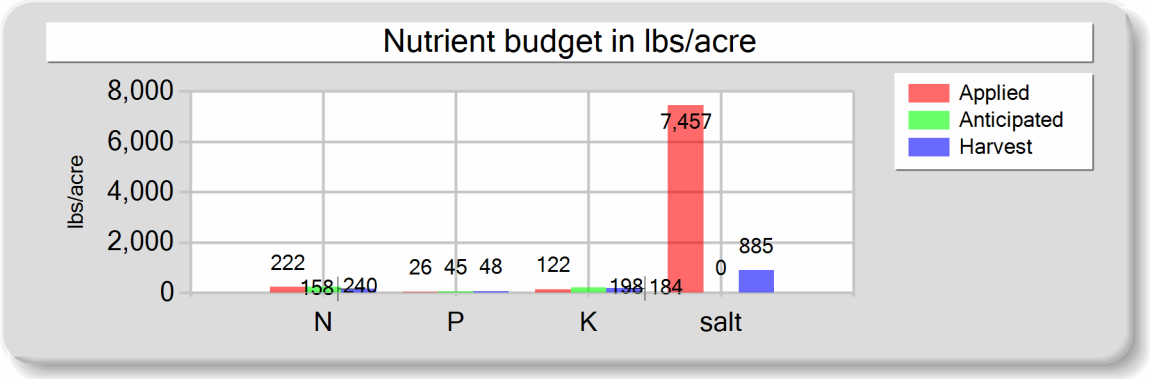
Fresh water applied	
15,959,725.00	*gallons*
587.74	*acre-inches*
11.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-4A - 06/04/2023: Corn, silage

Field name: BFH-4A Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	159.30	26.01	122.46	6,621.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	55.95	0.00	0.00	835.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.25	26.01	122.46	7,457.17
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	158.24	47.64	183.76	884.76
Nutrient balance	64.02	-21.63	-61.30	6,572.41
Applied to removed ratio	1.40	0.55	0.67	8.43

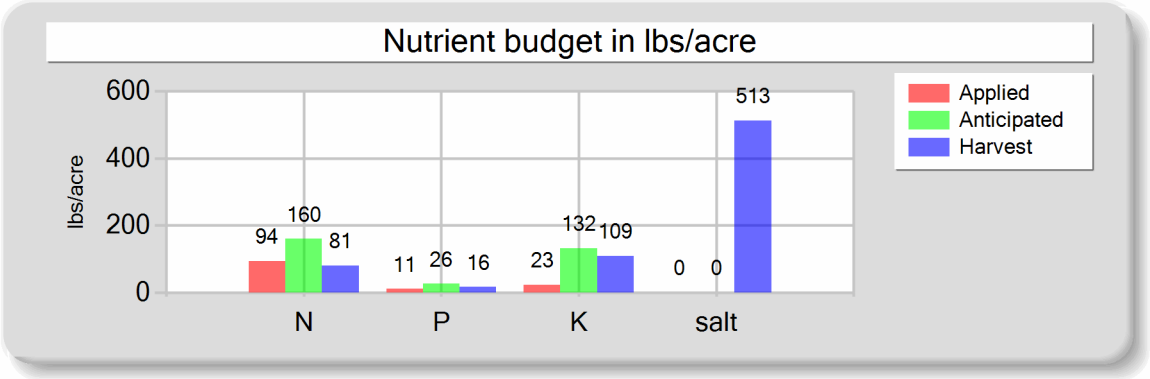
Fresh water applied	
65,061,613.00	*gallons*
2,396.00	*acre-inches*
46.98	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-4B - 11/22/2022: Oats, hay

Field name: BFH-4B Crop: Oats, hay Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	65.70	10.86	22.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	94.11	10.86	22.81	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	80.55	16.39	109.06	513.22
Nutrient balance	13.57	-5.53	-86.25	-513.22
Applied to removed ratio	1.17	0.66	0.21	0.00

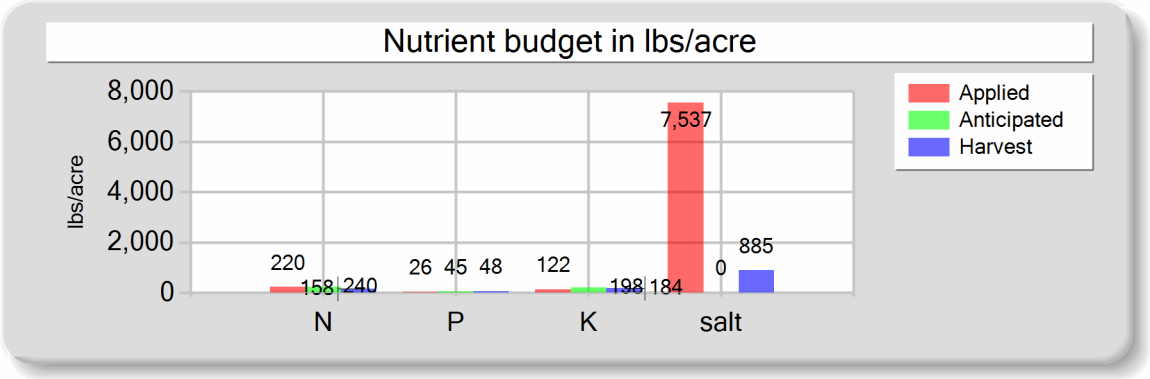
Fresh water applied	
15,959,725.00	*gallons*
587.74	*acre-inches*
11.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-4B - 06/04/2023: Corn, silage

Field name: BFH-4B Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	159.30	26.01	122.46	6,621.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	54.14	0.00	0.00	916.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.45	26.01	122.46	7,537.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	158.24	47.64	183.76	884.76
Nutrient balance	62.21	-21.63	-61.30	6,652.64
Applied to removed ratio	1.39	0.55	0.67	8.52

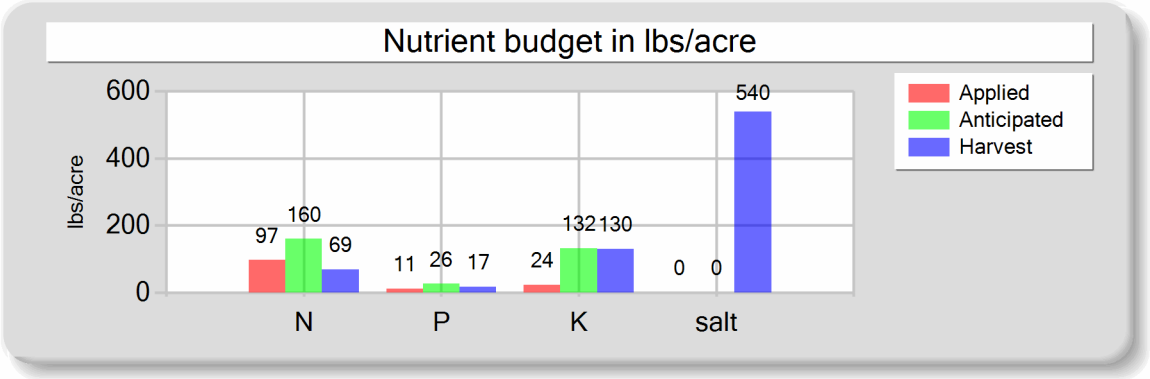
Fresh water applied	
65,957,464.00	*gallons*
2,428.99	*acre-inches*
47.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-5 - 10/28/2022: Oats, hay

Field name: BFH-5 Crop: Oats, hay Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	68.45	11.31	23.76	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21.41	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	96.86	11.31	23.76	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	69.19	17.39	130.24	540.20
Nutrient balance	27.67	-6.08	-106.48	-540.20
Applied to removed ratio	1.40	0.65	0.18	0.00

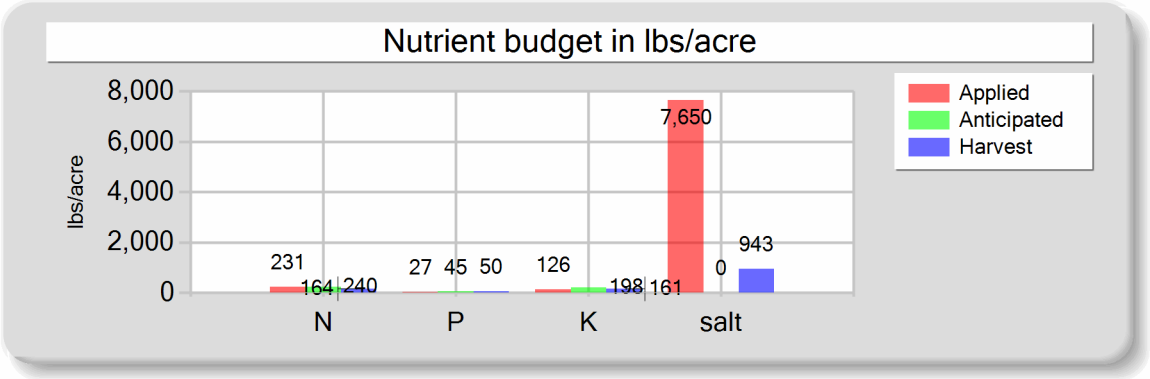
Fresh water applied	
36,300,551.00	*gallons*
1,336.83	*acre-inches*
11.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFH-5 - 06/04/2023: Corn, silage

Field name: BFH-5 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.99	26.77	126.06	6,816.23
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.80	0.00	0.00	833.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.79	26.77	126.06	7,650.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	164.31	50.21	161.27	943.28
Nutrient balance	66.48	-23.43	-35.21	6,706.81
Applied to removed ratio	1.40	0.53	0.78	8.11

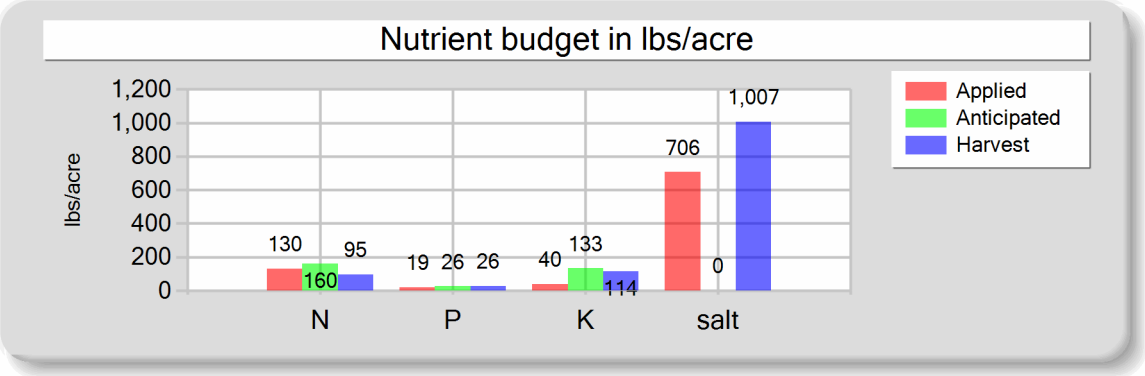
Fresh water applied	
154,381,515.00	*gallons*
5,685.35	*acre-inches*
49.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFHM-1 - 10/29/2022: Oats, silage-soft dough

Field name: BFHM-1 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	114.47	18.92	39.73	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.69	0.00	0.00	705.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	130.16	18.92	39.73	705.87
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	95.06	26.03	114.30	1,007.20
Nutrient balance	35.10	-7.11	-74.57	-301.33
Applied to removed ratio	1.37	0.73	0.35	0.70

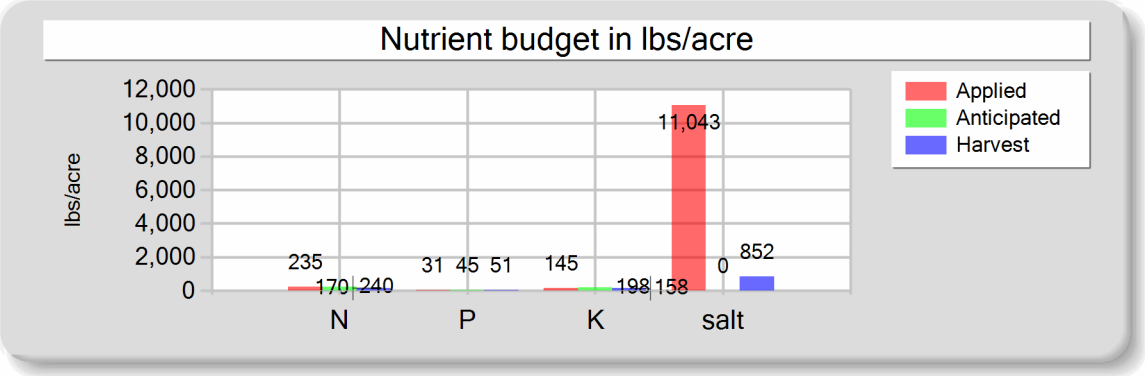
Fresh water applied	
6,232,653.00	*gallons*
229.53	*acre-inches*
10.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFHM-1 - 06/03/2023: Corn, silage

Field name: BFHM-1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	188.72	30.81	145.07	7,844.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.39	0.00	0.00	3,198.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.11	30.81	145.07	11,042.79
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	170.47	51.50	158.04	852.34
Nutrient balance	64.65	-20.68	-12.97	10,190.44
Applied to removed ratio	1.38	0.60	0.92	12.96

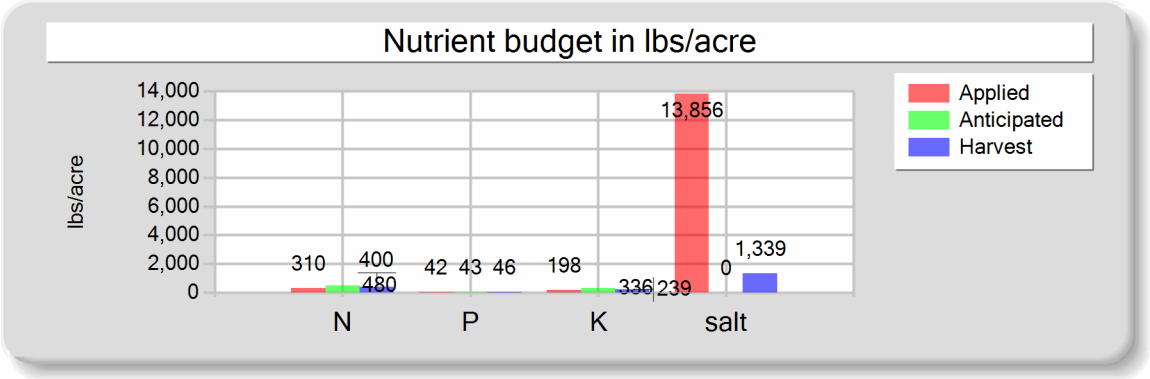
Fresh water applied	
28,243,791.00	*gallons*
1,040.12	*acre-inches*
49.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFHM-2 - 10/30/2022: Alfalfa, hay

Field name: BFHM-2 Crop: Alfalfa, hay Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	257.60	42.06	198.02	10,707.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	38.78	0.00	0.00	3,148.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	310.38	42.06	198.02	13,855.80
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	400.24	45.54	238.76	1,338.73
Nutrient balance	-89.86	-3.49	-40.74	12,517.07
Applied to removed ratio	0.78	0.92	0.83	10.35

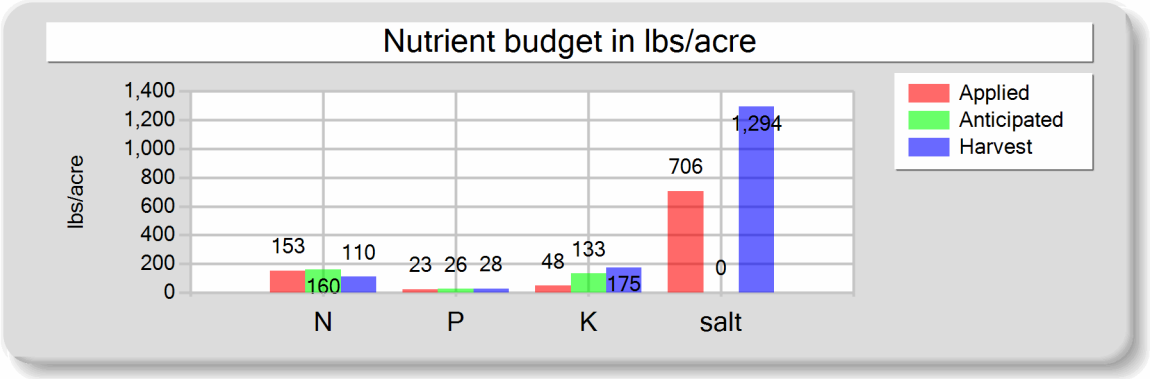
Fresh water applied	
35,744,953.00	*gallons*
1,316.37	*acre-inches*
48.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFHM-3 - 11/01/2022: Oats, silage-soft dough

Field name: BFHM-3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	137.77	22.77	47.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.69	0.00	0.00	705.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.46	22.77	47.82	705.87
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	110.29	27.57	174.98	1,293.80
Nutrient balance	43.17	-4.80	-127.16	-587.93
Applied to removed ratio	1.39	0.83	0.27	0.55

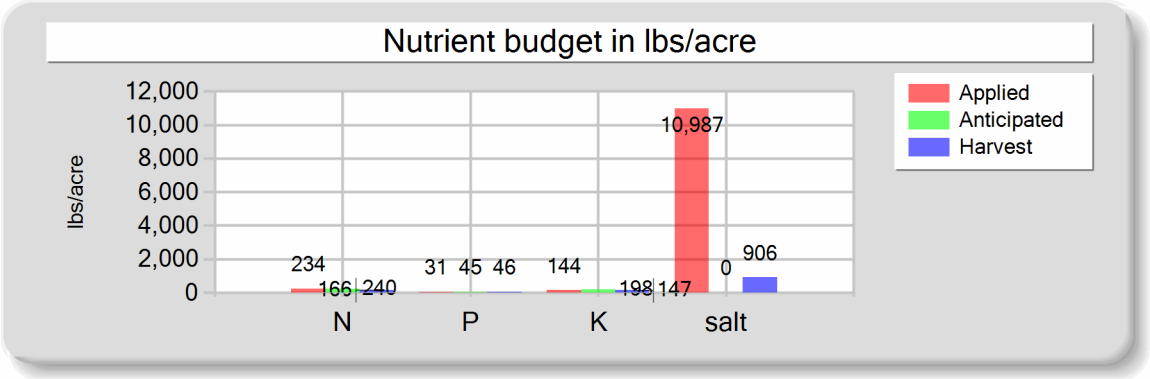
Fresh water applied	
27,304,956.00	*gallons*
1,005.55	*acre-inches*
10.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

BFHM-3 - 06/04/2023: Corn, silage

Field name: BFHM-3 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.39	30.59	144.05	7,788.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.39	0.00	0.00	3,198.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.78	30.59	144.05	10,987.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.45	46.14	146.67	906.40
Nutrient balance	67.33	-15.55	-2.63	10,080.96
Applied to removed ratio	1.40	0.66	0.98	12.12

Fresh water applied	
123,734,704.00	*gallons*
4,556.73	*acre-inches*
49.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure

Sample and source description: Separator Manure

Sample date: 10/05/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	4,000.00	8,400.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 12/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,400.00	11,300.00	6,700.00	6,500.00	3,000.00	2,200.00	11,000.00		61.10
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	130.00	104.00	0.00	2.20	19.50	402.00								3,790.00	1,950
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	570.00	438.00	0.00	1.40	75.00	725.00								8,340.00	4,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	528.00	387.00	0.00	1.00	87.20	694.00								8,200.00	3,830
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	551.00	458.00	0.00	1.00	78.70	839.00								8,880.00	4,150
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	466.00	446.00	0.00	3.00	119.00	800.00								8,570.00	4,980
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW #1 S.MB

Sample description: DW #1 S.MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.00	0.00	3.00								550.00	0
DL	0.01	0.01	0.01								10.00	1

DW #2 N. MB

Sample description: DW #2 N. MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.30	0.00	1.30								459.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW #4

Sample description: DW #4

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.80	0.00	18.80								1,830.00	0
DL	0.01	0.01	0.01								10.00	1

Drain Water

Drain Water

Sample description: Drain Water

Sample date: 06/17/2021 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.51	0.00	0.00								501.00	285
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

DW/IW

Sample description: DW/IW

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	32.90	0.00	32.90								2,020.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW Hunt

Sample description: IW Hunt

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.20	0.00	8.20								1,000.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

BF-1 - 10/29/2022: Oats, silage-soft dough

Oats BF-#1

Sample and source description: Oats BF-#1

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,200.00	2,000.00	13,800.00		9.60
DL	0.01	0.01	0.01		1.00

BF-1 - 06/03/2023: Corn, silage

BF-1 - 06/03/2023: Corn, silage

Corn BF-1

Sample and source description: Corn BF-1

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,700.00	15,600.00		8.40
DL	0.01	0.01	0.01		1.00

BF-2 - 10/30/2022: Oats, silage-soft dough

Oats BF-#2

Sample and source description: Oats BF-#2

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,600.00	1,700.00	13,500.00		8.90
DL	0.01	0.01	0.01		1.00

BF-2 - 06/04/2023: Corn, silage

Corn BF-2

Sample and source description: Corn BF-2

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	2,300.00	22,300.00		8.10
DL	0.01	0.01	0.01		1.00

BF-3 - 10/29/2022: Alfalfa, hay

BF 3 Alfalfa

Sample and source description: BF 3 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,000.00	2,900.00	33,800.00		11.20
DL	0.01	0.01	0.01		1.00

BF-4 - 10/10/2021: Alfalfa, hay

BF 4 Alfalfa

Sample and source description: BF 4 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,200.00	4,000.00	35,600.00		10.90
DL	0.01	0.01	0.01		1.00

BF-5 - 10/20/2017: Alfalfa, hay

BF 5 Alfalfa

Sample and source description: BF 5 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	4,200.00	22,400.00		10.00
DL	0.01	0.01	0.01		1.00

BF-7 - 10/20/2017: Alfalfa, hay

BF 7 Alfalfa

Sample and source description: BF 7 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,400.00	4,600.00	32,000.00		10.90
DL	0.01	0.01	0.01		1.00

BFH-1 - 11/18/2021: Alfalfa, hay

Alfalfa BFH 1

Sample and source description: Alfalfa BFH 1

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,900.00	3,100.00	18,000.00		8.80
DL	0.01	0.01	0.01		1.00

BFH 1 Alfalfa

Sample and source description: BFH 1 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,500.00	3,600.00	21,700.00		7.90
DL	0.01	0.01	0.01		1.00

BFH-2 - 11/18/2021: Alfalfa, hay

BFH-2 - 11/18/2021: Alfalfa, hay

Alfalfa BFH 2

Sample and source description: Alfalfa BFH 2

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,400.00	2,800.00	16,700.00		7.60
DL	0.01	0.01	0.01		1.00

BFH 2 Alfalfa

Sample and source description: BFH 2 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,500.00	3,200.00	16,100.00		7.10
DL	0.01	0.01	0.01		1.00

BFH-3 - 10/15/2019: Alfalfa, hay

Alfalfa BFH 3

Sample and source description: Alfalfa BFH 3

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,100.00	2,000.00	14,500.00		8.00
DL	0.01	0.01	0.01		1.00

BFH-3 - 10/15/2019: Alfalfa, hay

BFH 3 Alfalfa

Sample and source description: BFH 3 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,300.00	3,700.00	20,700.00		7.50
DL	0.01	0.01	0.01		1.00

BFH-4A - 11/07/2022: Oats, hay

Oat Hay BFH 4A

Sample and source description: Oat Hay BFH 4A

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,100.00	2,200.00	13,800.00		7.00
DL	0.01	0.01	0.01		1.00

BFH-4A - 06/04/2023: Corn, silage

Corn BFH-4

Sample and source description: Corn BFH-4

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	2,800.00	10,800.00		5.20
DL	0.01	0.01	0.01		1.00

BFH-4B - 11/22/2022: Oats, hay

Oat Hay BFH 4B

Sample and source description: Oat Hay BFH 4B

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,300.00	15,300.00		7.20
DL	0.01	0.01	0.01		1.00

BFH-4B - 06/04/2023: Corn, silage

Corn BFH-4

Sample and source description: Corn BFH-4

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	2,800.00	10,800.00		5.20
DL	0.01	0.01	0.01		1.00

BFH-5 - 10/28/2022: Oats, hay

Oat Hay BFH 5A

Sample and source description: Oat Hay BFH 5A

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,100.00	2,300.00	16,300.00		7.50
DL	0.01	0.01	0.01		1.00

BFH-5 - 10/28/2022: Oats, hay

Oat Hay BFH 5B

Sample and source description: Oat Hay BFH 5B

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	2,400.00	18,900.00		7.10
DL	0.01	0.01	0.01		1.00

BFH-5 - 06/04/2023: Corn, silage

Corn BFH-5

Sample and source description: Corn BFH-5

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	3,300.00	10,600.00		6.20
DL	0.01	0.01	0.01		1.00

BFHM-1 - 10/29/2022: Oats, silage-soft dough

Oats BFHm #1

Sample and source description: Oats BFHm #1

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,400.00	2,300.00	10,100.00		8.90
DL	0.01	0.01	0.01		1.00

BFHM-1 - 06/03/2023: Corn, silage

Corn BFHM-1

Sample and source description: Corn BFHM-1

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	2,900.00	8,900.00		4.80
DL	0.01	0.01	0.01		1.00

BFHM-2 - 10/30/2022: Alfalfa, hay

BFHM 2 Alfalfa

Sample and source description: BFHM 2 Alfalfa

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,000.00	3,300.00	17,300.00		9.70
DL	0.01	0.01	0.01		1.00

BFHM-3 - 11/01/2022: Oats, silage-soft dough

Oats BFHm #3

Sample and source description: Oats BFHm #3

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	2,600.00	16,500.00		12.20
DL	0.01	0.01	0.01		1.00

BFHM-3 - 06/04/2023: Corn, silage

Corn BFHM-3

Sample and source description: Corn BFHM-3

Sample date: 11/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,800.00	8,900.00		5.50
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

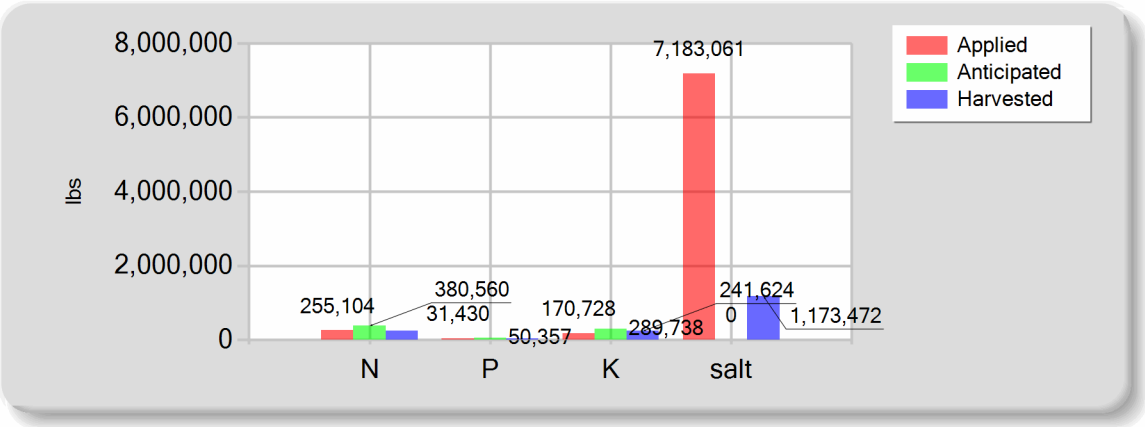
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

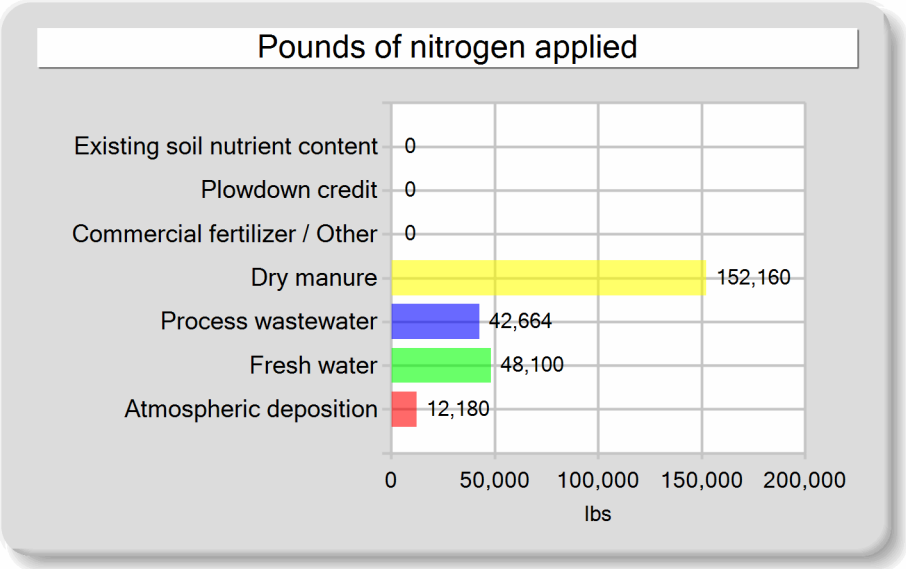
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

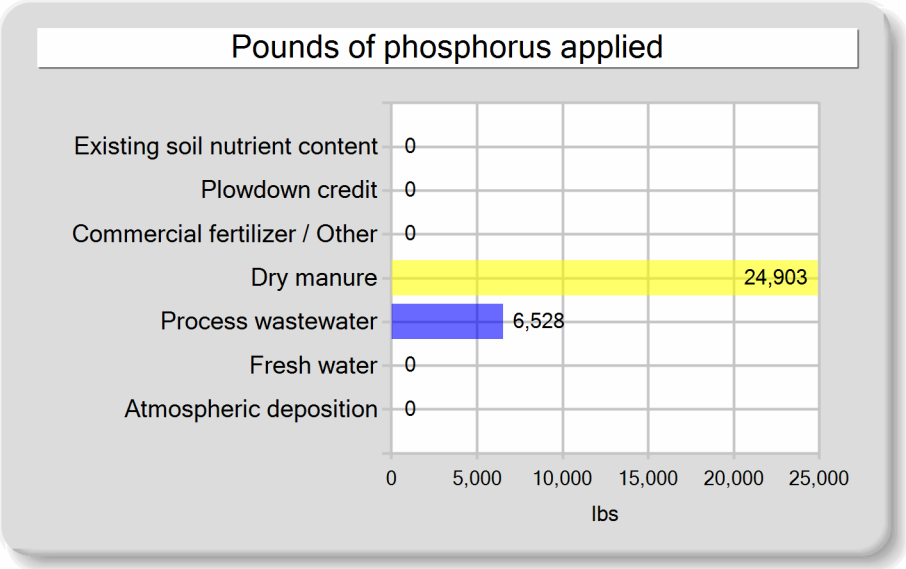
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152,160.13	24,902.62	104,436.87	5,089,197.41
Process wastewater	42,664.07	6,527.65	66,290.83	341,268.86
Fresh water	48,099.95	0.00	0.00	1,752,594.47
Atmospheric deposition	12,180.00	0.00	0.00	0.00
Total nutrients applied	255,104.15	31,430.27	170,727.70	7,183,060.74
Anticipated crop nutrient removal	380,560.00	50,357.40	289,738.00	0.00
Actual crop nutrient removal	244,079.50	41,478.83	241,624.26	1,173,472.07
Nutrient balance	11,024.65	-10,048.56	-70,896.56	6,009,588.66
Applied to removed ratio	1.05	0.76	0.71	6.12

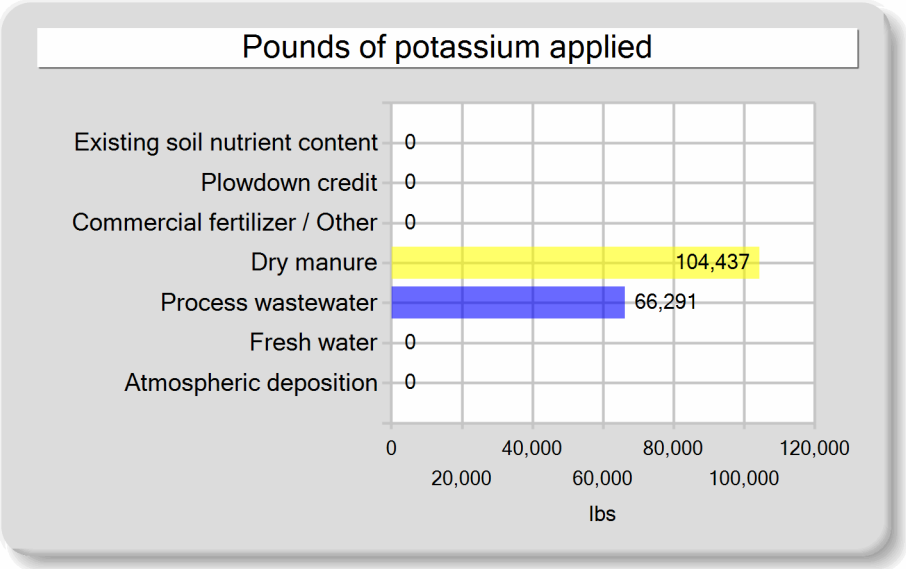
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

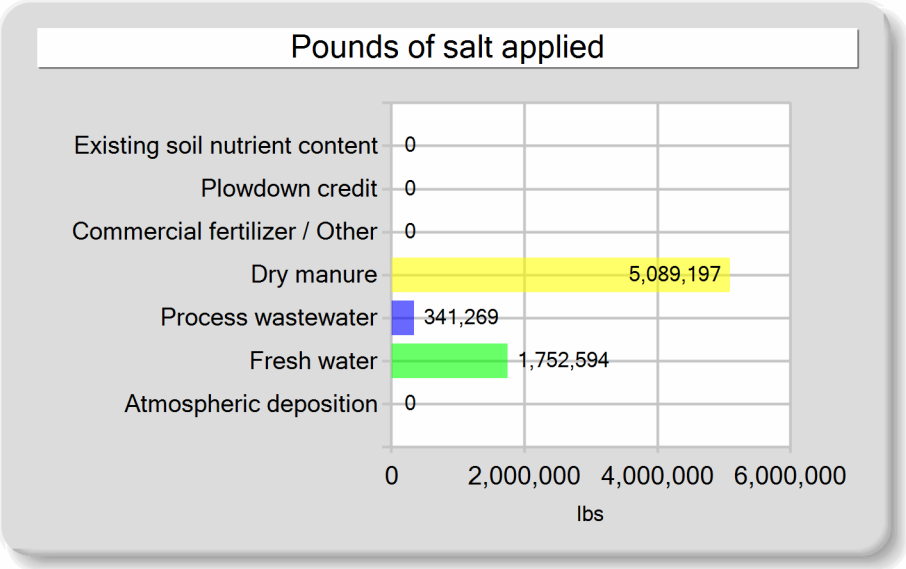


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Soares

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Soares Dairy Farms Inc.	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 2, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos CA 93635
00-0022799 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0989
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW #1 - S.MB	12/13/23	15:30	550	0.55	3.0	ND		7.8
2	DW #2 - N.MB	12/13/23	15:32	459	0.46	1.3	ND		8.0
3	DW #4	12/13/23	15:35	1830	1.83	18.8	ND		8.7
4	DW #3/IW (Bottle says "DW/IW")	12/13/23	15:35	2020	2.02	32.9	ND		7.3
5	IW Hunt	12/13/23	15:37	1000	1.00	8.2	ND		8.2

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos CA 93635
00-0022799 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0989
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW #1 - S.MB	12/13/23	15:30	550	0.55	**3.0**	8.19	0.68
2	DW #2 - N.MB	12/13/23	15:32	459	0.46	1.3	3.55	0.30
3	DW #4	12/13/23	15:35	**1830**	**1.83**	**18.8**	51.32	4.28
4	DW #3/IW (Bottle says "DW/IW")	12/13/23	15:35	**2020**	**2.02**	**32.9**	89.82	7.48
5	IW Hunt	12/13/23	15:37	**1000**	**1.00**	**8.2**	22.39	1.87

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0989-01	DW #1 - S.MB	Well Water	M Dekeyrel		12/13/2023 15:30
23L0989-02	DW #2 - N.MB	Well Water	M Dekeyrel		12/13/2023 15:32
23L0989-03	DW #4	Well Water	M Dekeyrel		12/13/2023 15:35
23L0989-04	DW #3/IW (Bottle says "DW/IW")	Well Water	M Dekeyrel		12/13/2023 15:35
23L0989-05	IW Hunt	Well Water	M Dekeyrel		12/13/2023 15:37

Default Cooler Temperature on Receipt °C: 5.0
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Sample Results

Sample: DW #1 - S.MB
23L0989-01 (Water)

Sampled: 12/13/2023 15:30
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		12/18/23 17:27	SM 2510 B		BEL0701
Electrical Conductivity umhos	**550**	umhos/cm	10.0	1		12/18/23 17:27	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:30	Field		BEL0631
Nitrate Nitrogen as NO3N	**3.0**	mg/L	0.1	1	10	12/17/23 02:54	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:27	SM 4500-H+	H	BEL0701
pH	**7.8**	units	1.0	1		12/18/23 17:27	SM 4500-H+	H	BEL0701

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Sample Results
(Continued)

Sample: DW #2 - N.MB
23L0989-02 (Water)

Sampled: 12/13/2023 15:32
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.46**	mmhos/cm	0.01	1		12/18/23 17:28	SM 2510 B		BEL0701
Electrical Conductivity umhos	**459**	umhos/cm	10.0	1		12/18/23 17:28	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:32	Field		BEL0631
Nitrate Nitrogen as NO3N	**1.3**	mg/L	0.1	1	10	12/17/23 03:14	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:28	SM 4500-H+	H	BEL0701
pH	**8.0**	units	1.0	1		12/18/23 17:28	SM 4500-H+	H	BEL0701

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Sample Results
(Continued)

Sample: DW #4
23L0989-03 (Water)

Sampled: 12/13/2023 15:35
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.83**	mmhos/cm	0.01	1		12/18/23 17:29	SM 2510 B		BEL0701
Electrical Conductivity umhos	**1830**	umhos/cm	10.0	1		12/18/23 17:29	SM 2510 B		BEL0701
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:35	Field		BEL0631
Nitrate Nitrogen as NO3N	**18.8**	mg/L	0.1	1	10	12/17/23 03:34	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:29	SM 4500-H+	H	BEL0701
pH	**8.7**	units	1.0	1		12/18/23 17:29	SM 4500-H+	H	BEL0701

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Sample Results
(Continued)

Sample: DW #3/IW (Bottle says "DW/IW")
23L0989-04 (Water)

Sampled: 12/13/2023 15:35
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.02**	mmhos/cm	0.01	1		12/18/23 17:45	SM 2510 B		BEL0704
Electrical Conductivity umhos	**2020**	umhos/cm	10.0	1		12/18/23 17:45	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:35	Field		BEL0631
Nitrate Nitrogen as NO3N	**32.9**	mg/L	0.1	1	10	12/17/23 03:55	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:45	SM 4500-H+	H	BEL0704
pH	**7.3**	units	1.0	1		12/18/23 17:45	SM 4500-H+	H	BEL0704

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Sample Results
(Continued)

Sample: IW Hunt
23L0989-05 (Water)

Sampled: 12/13/2023 15:37
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.00**	mmhos/cm	0.01	1		12/18/23 17:47	SM 2510 B		BEL0704
Electrical Conductivity umhos	**1000**	umhos/cm	10.0	1		12/18/23 17:47	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:37	Field		BEL0631
Nitrate Nitrogen as NO3N	**8.2**	mg/L	0.1	1	10	12/17/23 07:18	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:47	SM 4500-H+	H	BEL0704
pH	**8.2**	units	1.0	1		12/18/23 17:47	SM 4500-H+	H	BEL0704

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665										
Blank (BEL0665-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0665-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.2	90-110		
LCS (BEL0665-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
LCS (BEL0665-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.0	90-110		
LCS (BEL0665-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEL0665-DUP1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.639	10
Duplicate (BEL0665-DUP2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			0.787	10
Duplicate (BEL0665-DUP3)		**Source: 23L1000-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.612	10
Duplicate (BEL0665-DUP4)		**Source: 23L1005-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.7			0.941	10
Matrix Spike (BEL0665-MS1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L	5.000	5.8	102	90-110		
Matrix Spike (BEL0665-MS2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.3	102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665 (Continued)										
Matrix Spike (BEL0665-MS3)			**Source: 23L1000-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.2	91.6	90-110		
Matrix Spike (BEL0665-MS4)			**Source: 23L1005-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.7	99.3	90-110		
Reference (BEL0665-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0665-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0665-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701										
Blank (BEL0701-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0701-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0701-BLK3)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	8.1		1.0	units						
Duplicate (BEL0701-DUP1)		**Source: 23L0985-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.51		0.01	mmhos/cm		1.48			2.19	10
Electrical Conductivity umhos	1510		10.0	umhos/cm		1480			2.19	10
pH	7.7		1.0	units		7.7			0.130	10
Duplicate (BEL0701-DUP2)		**Source: 23L0989-03**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.84		0.01	mmhos/cm		1.83			0.730	10
pH	8.7		1.0	units		8.7			0.00	10
Electrical Conductivity umhos	1840		10.0	umhos/cm		1830			0.730	10
Reference (BEL0701-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	424			umhos/cm	426.0		99.6	90-110		
Reference (BEL0701-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0701-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0701-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0701 (Continued)										
Reference (BEL0701-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0701-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0701-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704										
Blank (BEL0704-BLK1)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0704-BLK2)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0704-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0704-DUP1)		**Source: 23L0999-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.01		0.01	mmhos/cm		1.01			0.248	10
Electrical Conductivity umhos	1010		10.0	umhos/cm		1010			0.248	10
pH	7.7		1.0	units		7.6			0.784	10
Duplicate (BEL0704-DUP2)		**Source: 23L1004-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.11		0.01	mmhos/cm		1.11			0.108	10
pH	7.8		1.0	units		7.7			0.775	10
Electrical Conductivity umhos	1110		10.0	umhos/cm		1110			0.108	10
Reference (BEL0704-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	426			umhos/cm	426.0		99.9	90-110		
Reference (BEL0704-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0704-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Soares Badger Flat Dairy
14515 Badger Flat Rd
Los Banos, CA 93635

Account# 00-0022799
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:01

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704 (Continued)										
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0704-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/14/23 14:20 23L0989

MMM

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 22799 | 8

Purchase Order No. _____ Results Needed By _____

Client: Soares - Badger Flat Dairy
Address: 14515 Badger Flat Road
City, State, Zip: Los Banos, CA 93635
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by _____

Ranch _____

Date sampled: 12/13/23

Sampled by: M Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 5 No. Bottles: 5

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES	Sampled From:	Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW#1-S.MB		12/13	3:30	Ø	5.0
2. DW#2-N.MB		12/13	3:32	Ø	4.9
3. DW#4		12/13	3:35	Ø	2.2
4. DW#1W (bottle says "DW#1W")		12/13	3:35	Ø	4.3
5. IW Hunt	MJS 12/14/23	12/13	3:37	Ø	5.0
6.					
7.					
8.					
9.					
10.					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeyrel	CAS inc	12/13/23 @ 3:38pm	12/14/23 12p
Second	CSM	DLI	12/14/23 12 pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1	1					
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L0989

SENDOUT